Pranav Khurana

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EDUCATION

University of Illinois Urbana-Champaign **Expected May 2030**

Bachelor of Science, Computer Engineering

- **Achievements:** Red Hat Technology Scholarship Award
- **Coursework:** ECE 120 Introduction to Computing (digital logic, C programming), Data Structures and Algorithms (UMass Dartmouth), Statistics

SKILLS

Languages: C/C++, Python, Java
Software & Frameworks: PyTorch, OpenCV, YOLO, ROS, Linux, Git, Multithreaded Programming
Hardware: Embedded Systems, Raspberry Pi, Arduino, Autodesk Fusion 360 (CAD), CNC Machining, Soldering

EXPERIENCE

National Organization for Business and Engineering (NOBE) | ***Product Developer*** **Sep 2026 - Present**

- Building an interceptor drone on a project team reporting to client Haylon Technologies; designed the full drone assembly in Autodesk Fusion 360
- Implemented a YOLO26 computer-vision pipeline that tracks a target and computes the intercept path, including distance, bearing, and time to impact
- Extended tracking to moving targets by estimating target velocity and accounting for it in the intercept calculation

Syntropy Systems (Startup) | ***Technical Intern*** **Jun 2026 - Aug 2026**

- Implemented neural network architectures from scratch in Python and PyTorch
- Applied KV caching and speculative decoding to reduce LLM generation latency for agents that automate tasks in Slack channels

Ghost Robotics, FIRST Tech Challenge Team 3565 | ***Team Captain & Tech Lead*** **Sep 2022 - Aug 2026**

- Finished 3rd Alliance in the World out of over 8,000 registered teams
- Led development of Java-based computer-vision pipelines using OpenCV for autonomous goal aiming, increasing the robot's targeting reliability during matches
- Improved shooting accuracy by 35% by fusing positional data from AprilTags with dead-wheel odometry
- Spearheaded outreach events and summer camps that raised over $50,000 for the robotics program

Worcester Polytechnic Institute BASH Lab | ***Research Assistant*** **May 2025 - Sep 2025**

- Developed an embedded earbud platform with Python and Raspberry Pi on a Linux workstation, then open-sourced the hardware and software so researchers can replicate the physiological-data collection system
- Implemented LMS filters to reduce noise by over 60%, isolating physiological signals such as breathing from raw sensor data
- Wired a custom stereo-microphone circuit and configured Raspberry Pi hardware to collect internal and external audio signals

Westford Academy | ***AI Club Founder & President*** **Sep 2024 - Jun 2025**

- Founded and grew a 15-member club, teaching gradient descent, neural networks, and backpropagation and guiding students in building machine-learning models on Raspberry Pi

UMass Lowell PeARL Lab | ***Research Assistant*** **May 2023 - Jul 2023**

- Developed an efficient occlusion-handling method for multi-camera systems and optimized camera role assignment using Stackelberg game theory, improving tracking accuracy by up to 20% over Q-learning and DQN baselines
- Configured servos using ROS scripts to track a partially or fully occluded face, and collected facial-recognition consistency data

PROJECTS

Autonomous Stabilizing Drone **Jun 2024 - Oct 2024**

- Built and programmed a fully functional embedded system by wiring motors and a microcontroller onto a store-bought frame
- Used a six-axis gyroscope and Kalman filtering to obtain stable position data, then implemented automatic stabilization with three concurrent feedback systems using multithreaded programming

AI Smart Traffic Light **Jun 2024 - Feb 2025**

- Programmed a computer vision and machine learning simulator in Python to model and optimize traffic flow at a three-way intersection, training models with the NEAT library on real-time vehicle data
- Built physical prototypes using Google Coral Accelerator and Raspberry Pi, and coordinated with highway facilities after simulations showed a 2.4x decrease in waiting time

PUBLICATIONS

- Osooli, Joshi, **Khurana**, et al. “Releaf: An Efficient Method for Real-Time Occlusion Handling by Game Theory,” *Sensors*, 2024
- Co-authored paper and poster on an open-source earbud platform, IEEE MIT Undergraduate Research Technology Conference